

Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Fire Station San Luis, Pampanga

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BACKGROUND OF THE STUDY

Fire incidents pose significant threats to lives, properties, and communities, making timely reporting and efficient emergency response essential. However, traditional fire incident reporting and record management often rely on manual processes, which may lead to delays, redundant records, difficulties in retrieving information, and limited coordination during emergency operations. These challenges can affect the efficiency of the Bureau of Fire Protection (BFP) in responding to emergencies.

Advancements in web, mobile, and Artificial Intelligence (AI) technologies provide opportunities to improve emergency response management. Through an integrated digital platform, residents can report fire incidents in real time, while BFP personnel can receive notifications, verify reports, monitor incidents, and manage emergency operations more efficiently. AI can further assist by analyzing incident severity, location, potential risk, and available response resources to recommend response priorities, supporting faster and more informed decision-making.

To address these challenges, this study proposes the development of an Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Bureau of Fire Protection – Fire Station San Luis, Pampanga. The proposed web and mobile application aim to improve fire incident reporting, enhance emergency response coordination, and support timely decision-making through AI-assisted incident prioritization, real-time notifications, interactive mapping, and centralized incident management. Ultimately, the system seeks to improve the efficiency and effectiveness of fire emergency services in the Municipality of San Luis, Pampanga.

PROBLEM STATEMENT

This study aims to answer the following questions:

General Problem

How can an Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization improve fire incident reporting, emergency response coordination, and incident management for the Bureau of Fire Protection – Fire Station San Luis, Pampanga?

Specific Problems

1. How can the system improve the reporting, recording, and retrieval of fire incident information to reduce manual documentation, reporting delays, and data redundancy?
2. How can the system provide real-time monitoring and tracking of reported fire incidents to enhance emergency response coordination and incident management?
3. How can the system utilize Artificial Intelligence to automatically prioritize reported fire incidents based on factors such as incident severity, location, potential risk, and available response resources?
4. How can the system enable citizens to report fire incidents and allow Bureau of Fire Protection personnel to receive, verify, update, and manage incident information through web and mobile applications?
5. How can the system provide automated notifications, response status tracking, and analytical dashboards to assist Bureau of Fire Protection personnel in making timely decisions and improving emergency response operations?

OBJECTIVES

General Objective

To develop an Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Bureau of Fire Protection – Fire Station San Luis, Pampanga that improves fire incident reporting, enhances emergency response coordination, and supports timely decision-making through intelligent incident analysis and prioritization.

Specific Objectives

Specifically, this study aims to:

1. To develop a web and mobile application that improves the reporting, recording, storage, and retrieval of fire incident information to reduce manual documentation, reporting delays, and data redundancy.
2. To provide an AI-powered incident prioritization system that analyzes reported fire incidents, evaluates factors such as severity, location, potential risk, and available response resources, and automatically prioritizes emergency response operations.
3. To implement an automated incident management and reporting module that generates fire incident reports, response records, and statistical summaries to improve the efficiency of Bureau of Fire Protection personnel.
4. To develop a synchronized web and mobile platform that enables citizens to report fire incidents and allows Bureau of Fire Protection personnel to receive, verify, update, and manage incident information in real time.
5. To integrate AI-based analytics, automated notifications, response status tracking, and interactive dashboards that assist Bureau of Fire Protection personnel in making timely and informed emergency response decisions.

TARGET USERS

Primary Users

Bureau of Fire Protection (BFP) Personnel – Fire Station San Luis, Pampanga

- Receive and manage fire incident reports submitted through the web and mobile applications.
- Verify, update, and monitor the status of reported fire incidents.
- Utilize AI-based incident prioritization to determine the urgency of emergency response.
- Generate fire incident reports, response records, and operational statistics.
- Monitor response activities and coordinate emergency operations.

Fire Station Officer / Station Commander

- Oversee fire incident response operations and personnel deployment.
- Review AI-generated incident priorities, response analytics, and operational dashboards.
- Monitor response performance and resource utilization.
- Generate operational and incident summary reports.
- Support strategic planning and decision-making for fire prevention and emergency response.

Emergency Response Personnel (Firefighters)

- Receive real-time notifications of assigned fire incidents.
- View incident details, locations, and AI-assigned priority levels.
- Update the status of emergency response operations through the mobile application.
- Record response actions and incident outcomes.
- Improve coordination and communication during emergency response.

Secondary Users

Residents of the Municipality of San Luis

- Report fire incidents through the web or mobile application.
- Receive updates on the status of submitted incident reports.
- Access emergency contact information and fire safety advisories.
- Benefit from faster and more efficient emergency response services.

Municipal Disaster Risk Reduction and Management Office (MDRRMO)

- Access summarized fire incident statistics and emergency response reports.
- Monitor fire incident trends to support disaster preparedness and risk reduction initiatives.

- Coordinate with the Bureau of Fire Protection during emergency situations.
- Support planning, resource allocation, and emergency management activities.

Local Government Unit (LGU) of San Luis

- Review summarized fire incident reports and statistical data.
- Monitor fire-related incidents within the municipality.
- Support policy development, fire prevention programs, and resource allocation.
- Utilize system-generated reports for planning and decision-making.

PROPOSED TECHNOLOGIES

The proposed Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Bureau of Fire Protection – Fire Station San Luis, Pampanga will be developed as a web and mobile application to provide efficient fire incident reporting, emergency response coordination, incident management, and AI-assisted decision support.

A. Software Technologies

Development Tools

- Visual Studio Code
- Android Studio
- XAMPP
- Git and GitHub
- Postman
- Google Chrome / Microsoft Edge

Frontend

- HTML5

- CSS3
- JavaScript
- Bootstrap 5
- Chart.js
- Leaflet.js (for interactive incident mapping)

Backend

- Python
- Django
- Django REST Framework (DRF)
- REST API

Mobile Application

- Flutter
- Dart

AI Technologies

- Scikit-learn (AI-Based Incident Prioritization)
- Pandas (Data Processing and Analysis)
- NumPy (Numerical Computing)
- OpenAI API or Gemini API (*Optional – for AI-assisted report summarization and intelligent recommendations*)

Database

- MySQL

Mapping and Location Services

- Google Maps API or OpenStreetMap

- Geolocation API

Notification Services

- Firebase Cloud Messaging (FCM)

Operating Systems

- Windows 11
- Android
- Windows Server or Linux Server

B. Hardware

Development Hardware

- Laptop or Desktop Computer
 - Intel Core i5 processor or higher
 - 8 GB RAM or higher
 - 256 GB SSD or higher

Deployment Hardware

- Desktop computers for the Bureau of Fire Protection – Fire Station San Luis, Pampanga
- Android smartphones or tablets for BFP personnel and firefighters
- Android smartphones for residents (for mobile fire incident reporting)
- Printer
- Stable internet connection
- Wi-Fi Router / Local Area Network (LAN)

INNOVATION

The proposed Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Fire Station San Luis, Pampanga introduces an integrated web and mobile application designed to improve fire incident reporting, emergency response coordination, and incident management. Unlike traditional manual reporting and record-keeping processes, the system enables real-time fire incident reporting, centralized incident record management, and faster communication between residents and Fire Station San Luis personnel.

The system's key innovation is its AI-based incident prioritization feature, which analyzes reported fire incidents using factors such as incident severity, location, potential risk, and available response resources to automatically assign response priority. This assists emergency personnel in identifying high-priority incidents and allocating resources more effectively during multiple or simultaneous emergencies. Residents can report fire incidents through the mobile application, while fire personnel can receive notifications, verify reports, monitor incident status, and manage response operations through a web-based dashboard. The integration of AI-powered prioritization, interactive incident mapping, automated notifications, and analytical dashboards improves response efficiency, supports timely decision-making, and enhances the overall delivery of fire emergency services in the Municipality of San Luis, Pampanga.

FEASIBILITY

Technical Feasibility

The proposed system is technically feasible because it will utilize widely used and reliable technologies such as Flutter, Django, MySQL, Django REST Framework (DRF), and Scikit-learn. These technologies support the development of secure, scalable, and responsive web and mobile applications. The integration of AI-based incident prioritization can be implemented using machine learning algorithms that analyze incident information and assist fire personnel in determining the urgency of emergency response.

Operational Feasibility

The system is operationally feasible as it is designed to simplify the daily tasks of Fire Station San Luis personnel by streamlining fire incident reporting, emergency response coordination,

incident monitoring, and report generation. It enables residents to report fire incidents through a mobile application while allowing fire personnel to verify, manage, and monitor incidents using a centralized web-based system. The AI-based incident prioritization feature supports faster and more informed decision-making, improving the overall efficiency of emergency response operations.

Economic Feasibility

The project is economically feasible because it utilizes open-source software, frameworks, and development tools, significantly reducing development costs. Existing desktop computers at the Fire Station and Android smartphones used by fire personnel can be utilized for system deployment, minimizing the need for additional hardware investment. Only a stable internet connection and basic networking equipment are required to support system operation.

Schedule Feasibility

The development of the system is feasible within the allotted capstone timeline. The project will follow a structured software development process, including planning, requirements analysis, system design, development, testing, implementation, and evaluation, to ensure the successful completion of the proposed Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Fire Station San Luis, Pampanga within the prescribed schedule.

SCOPE AND LIMITATIONS

Scope

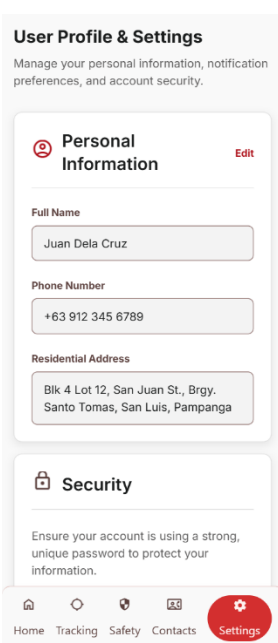
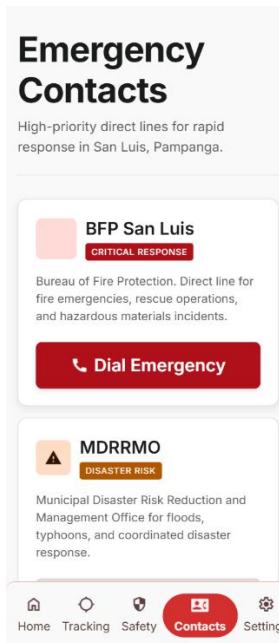
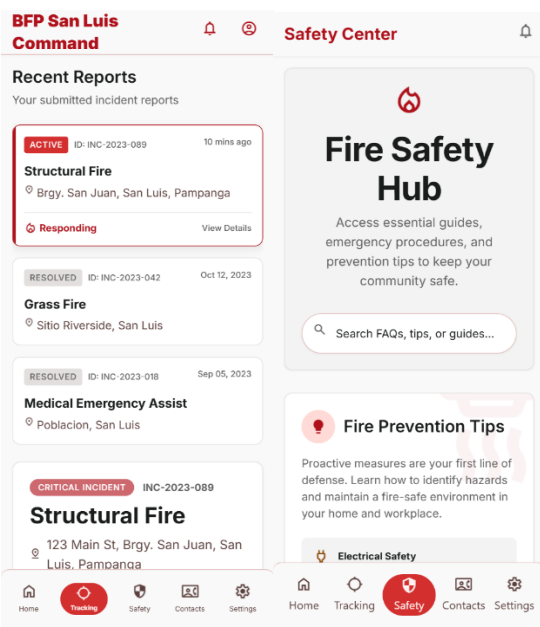
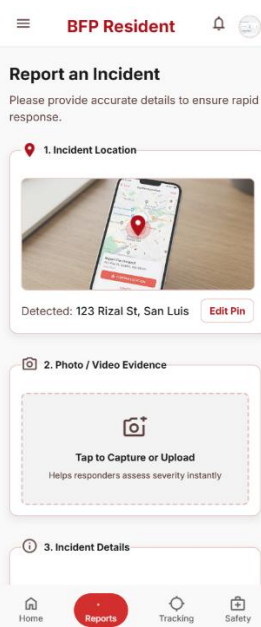
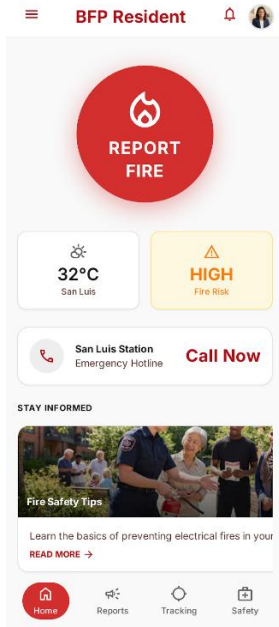
The proposed Intelligent Fire Incident Reporting and Emergency Response Management System with AI-Based Incident Prioritization for the Fire Station San Luis, Pampanga is a web and mobile application designed to improve fire incident reporting, emergency response coordination, and incident management. The system will allow residents to report fire incidents through the mobile application by providing incident details, location, and supporting images. Authorized Fire Station personnel will use the web application to receive, verify, manage, and monitor reported incidents, assign response teams, generate incident reports, and view

operational statistics through a centralized dashboard. The system will also integrate an AI-based incident prioritization feature that analyzes reported incident information and recommends response priority levels based on factors such as incident severity, location, potential risk, and available response resources. Additionally, the system will provide real-time notifications, interactive incident mapping, response status tracking, and analytical dashboards to support efficient emergency response operations.

Limitations

The system is intended for use only by the Fire Station San Luis, Pampanga and its authorized personnel. The AI-based incident prioritization feature serves only as a decision-support tool and does not replace the professional judgment or operational decisions of fire officers during emergency response. The accuracy of the system's prioritization depends on the completeness and accuracy of the information submitted by users. The system requires an internet connection for real-time reporting, notifications, and synchronization between the web and mobile applications. It is limited to managing and monitoring reported fire incidents and does not include advanced emergency dispatch systems, GPS vehicle tracking, integration with other government emergency agencies, or predictive fire forecasting beyond the features defined within the scope of this study.

EXPECTED OUTPUT



BFP San Luis

Command Center

New Incident Report

Dashboard

Personnel

Duty Roster

Incident Logs

Resource Tracking

Analytics

System Settings

Logout

Personnel Command

Manage deployment, rosters, and operational status.

Active Directory

Duty Roster

All Units

All Statuses

Filter

Export

Personnel Name	Rank / ID	Assigned Unit	Current Status	Actions
 Mateo Santos	SFO4 ID: BFP-9902	Rescue 2 (Alpha Shift)	DISPATCHED	
 Capt. Aris Delgado	Station Commander ID: BFP-8811	Command Staff	ON DUTY	
 Clara Reyes	FO1 ID: BFP-0145	Engine 1 (Bravo Shift)	OFF DUTY	
 Julio Fernandez	FO2 ID: BFP-7732	Engine 2	TRAINING	

Showing 1 to 4 of 48 personnel

BFP San Luis

Command Center

New Incident Report

Dashboard

Personnel

Duty Roster

Incident Logs

Resource Tracking

Analytics

System Settings

Logout

Personnel

Reports

Analytics

Settings

Emergency Dispatch

Reports & Analytics

Command level oversight and performance metrics.

Last 30 Days

Export

Target: < 5m

Avg Response Time

4m 12s

↓ 12%

Total Incidents

143

↑ 5%

Resolution Rate

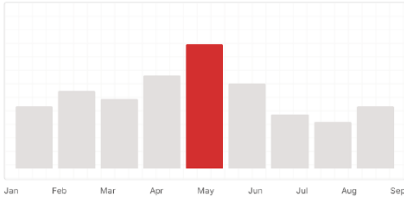
98.5%

Personnel on Duty

24 / 32

Active

Response Time Trends



Month	Response Time
Jan	~4m 15s
Feb	~4m 10s
Mar	~4m 12s
Apr	~4m 08s
May	~4m 30s
Jun	~4m 10s
Jul	~4m 05s
Aug	~4m 08s
Sep	~4m 10s

Hotspots

San Isidro	42 Incidents
Sto. Tomas	28 Incidents
San Juan	15 Incidents
San Nicolas	12 Incidents